

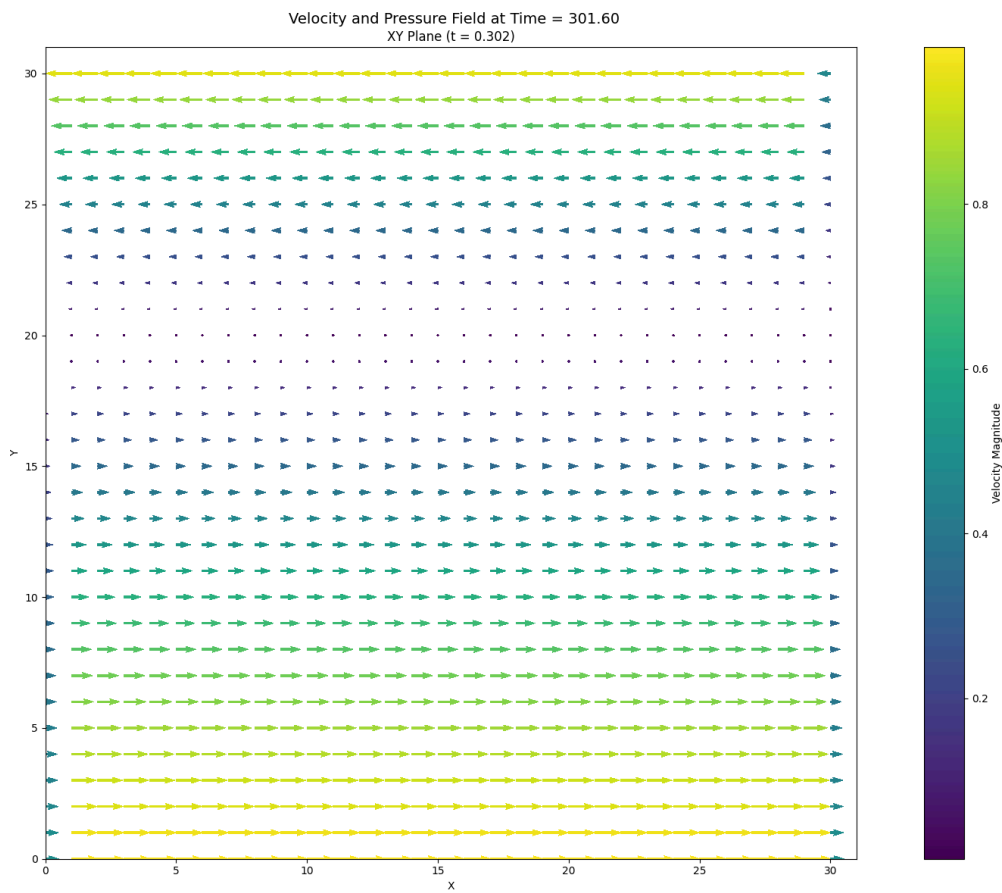
Experimental Report

Case: Poiseuille Flow with Moving Walls

Description of Flow:

This case represents pressure-driven channel flow in which the two horizontal walls move at a prescribed speed while the fluid is forced along the streamwise direction by a constant body force. The boundary motion adds wall shear to the conventional Poiseuille profile, producing a skewed velocity distribution that is still dominated by viscous diffusion and the imposed forcing. The target is to recover the governing streamwise momentum equation from the saved velocity and pressure fields.

Numerical identification of the streamwise momentum equation from saved velocity and pressure fields.



1. Governing Equation Verification

Reference equation

$$u_t = 8.80434783e-05 u_{yy} + 9.81000000e+00 \text{one}$$

Learned equation

$$u_t = -2.08702206e-10 u u_y + 2.68851484e-07 u + 1.56320160e-08 u_y + 8.84726794e-05 u_{yy} + 9.85782285e+00 \text{one}$$

Term	Reference coefficient	Learned coefficient	Absolute error
$u \cdot u_y$	$+0.00000000e+00$	$-2.08702206e-10$	$2.08702206e-10$
$u \cdot u_{yy}$	$+0.00000000e+00$	$+4.66507141e-12$	$4.66507141e-12$
$u_y \cdot u_{yy}$	$+0.00000000e+00$	$+1.34869663e-13$	$1.34869663e-13$
u	$+0.00000000e+00$	$+2.68851484e-07$	$2.68851484e-07$
u_y	$+0.00000000e+00$	$+1.56320160e-08$	$1.56320160e-08$
u_{yy}	$+8.80434783e-05$	$+8.84726794e-05$	$4.29201163e-07$
1	$+9.81000000e+00$	$+9.85782285e+00$	$4.78228475e-02$

2. Simulation Parameters

Quantity	Symbol	Value	Unit
Fluid density	ρ	$9.20000000e+02$	kg m^{-3}
Dynamic viscosity	ν	$8.10000000e-02$	Pa s
Kinematic viscosity	ν	$8.80434783e-05$	$\text{m}^2 \text{s}^{-1}$
Characteristic velocity	u_{max}	$1.00000000e+00$	m s^{-1}
Characteristic length	D	$5.00000000e-03$	m
Reynolds number	Re	$1.97740055e+01$	dimensionless
Body force	G	$9.81000000e+00$	m s^{-2}
Grid points	N_x	$3.10000000e+01$	points
Unused transverse grid points	N_y	$3.10000000e+01$	points
Grid spacing	Δy	$1.61290323e-04$	m
Time step	Δt	$1.31646880e-04$	s
Numerical threshold	ϵ	$1.00000000e-15$	-

3. Numerical Discretisation

Quantity	Symbol	Value	Unit
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Parameters listed above are the values used by the selected solver.

4. Data and Pre-processing

Quantity	Value
Loaded field shape (t, y)	$(2142, 31)$
Regression field shape (t, y)	$(542, 27)$

Quantity	Value
Regression samples	14,634
Dictionary terms	7
First saved time index	150
Temporal crop at each boundary	800 layers
x-boundary crop	0 points
y-boundary crop	2 points

5. STRidge Selection and Error Metrics

Metric	Value
Regularisation parameter, λ	1.00000000e-09
Selected tolerance	1.00000000e-06
L0 penalty	1.00000000e-06
Training RMSE	2.63621763e-09
Validation RMSE	3.34331075e-09